

Amit Kumar Mohanty

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SUMMARY

Machine Learning Engineer with ~1 year of experience building, training, and deploying production-grade AI/ML systems, specialized in LLMs, Agentic AI, deep learning, and end-to-end ML pipelines. Recognized as "Rookie of the Year 2025" at Keywords Studios for exceeding all KPIs. Skilled in Python, TensorFlow/PyTorch, fine-tuning (LoRA/QLoRA), RAG, and cloud ML (AWS, GCP Vertex AI), with a track record of taking models from research to scalable production and a peer-reviewed publication in computer vision.

EDUCATION

K.R. Mangalam University

July 2021 – July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML), Gurgaon | Cumulative GPA: 8.5/10

Relevant Coursework: Data Structures & Algorithms, NLP, Computer Vision, DBMS, Machine Learning, Deep Learning

EXPERIENCE

Research Associate – AI, Keywords Studios India – Gurgaon, Haryana

May 2025 – Present

- Trained and optimized enterprise-grade Agentic AI models to autonomously execute complex multi-step workflows (flight booking, e-commerce ordering, web data retrieval), improving task-completion reliability.
- Built custom data pipelines and delivered 85% of self-generated training datasets directly to production as the first member of a global team, accelerating model readiness.
- Awarded "Rookie of the Year 2025" for surpassing all performance KPIs through advanced model training, data-pipeline optimization, and cross-functional collaboration.

Artificial Intelligence Intern, Infosys – Remote

November 2024 – February 2025

- Engineered an AI call-center system with LangChain and LangSmith, boosting model response accuracy by 35% and elevating customer-AI interaction quality.
- Developed and deployed LLM features that reduced system latency by 25%, streamlining the model deployment process.
- Collaborated with a team of 12 engineers to integrate end-to-end ML workflows and earned selection to present at the Best Project by Indian Interns initiative.

Data Analytics & ML Intern, QL2 Software – Gurgaon, Haryana

June 2024 – August 2024

- Built predictive ML models for competitive pricing intelligence and performed large-scale data extraction and analysis to deliver business insights.

PROJECTS

Dynamic Screen Companion

June 2025 – Present

- Built a real-time multi-modal assistant integrating the Gemini API to analyze live screen activity and deliver continuous voice feedback and interactive support.
- Engineered a high-performance FastAPI model-serving backend handling concurrent inference streams, reducing manual task time by 40%.
- Designed the end-to-end multi-modal architecture from scratch and optimized OCR pipelines to maintain 95%+ extraction accuracy across varied screen layouts.

Diabetic Optiscan

January 2025 – June 2025

- Led integration of a Vision Transformer (ViT-B/16) model on the APTOS dataset, optimizing self-attention to capture complex spatial patterns in high-resolution retinal images.
- Designed a custom wavelet modification technique that boosted detection precision by 50% over legacy methods.
- Co-authored a peer-reviewed research paper detailing the system architecture and performance breakthroughs.

Currency Detection App for Visually Impaired

July 2024 – December 2024

- Trained an image-classification model on 200,000+ banknote images achieving 96% accuracy, deployed in a mobile app with voice feedback to help prevent fraud.
- Conducted user interviews with visually impaired individuals to refine real-world accessibility features.

SKILLS

Languages & Frameworks: Python, C++, SQL, R, FastAPI, TensorFlow, PyTorch, OpenCV, Scikit-learn, LangChain, LangGraph

AI & Machine Learning: LLMs, Generative AI, Agentic AI, RAG, Fine-tuning (LoRA, QLoRA), Deep Learning, Transformers, Hugging Face, BERT, NLP, Prompt Engineering, Model Evaluation & Optimization, OOP, DSA

MLOps & Deployment: Docker, Kubernetes, CI/CD, MLflow, Model Serving, Model Monitoring, Data Pipelines, Git, Vector Databases (FAISS/Chroma)

Data & Cloud Tools: AWS, GCP (Vertex AI), Power BI, MySQL, MongoDB, Pandas, NumPy, Statistical Analysis

CERTIFICATIONS

Ducat India – Data Science Professional Training

April 2024 – April 2025

Professional training in Data Science: Python, statistics, machine learning, data analysis, and predictive modeling.

AI/ML for Geodata Analysis

Offered by ISRO (Indian Space Research Organisation) – applying AI/ML to geospatial data analysis.